

EE 304 Embedded Systems – CAPSTONE DESIGN COMPETITION

1.0 GENERAL INSTRUCTIONS

- **Team Size:** Maximum **3 Students** per group.
 - **Weight:** **40%** of Final Grade.
 - **The "Split-Brain" Requirement:** Every project **MUST** use a **Raspberry Pi (4 or 5)** as the "Brain" (Vision/AI/Logic) and an **Arduino Uno R4 or Pico W** as the "Muscle" (Motor Control/Sensor Polling). They must communicate via **UART Serial**.
 - **The Competition:**
 - **Demo Day:** Week 13 & 14 (In-class).
 - **Voting:** Students will vote for the best systems based on Stability, Complexity, and Latency.
 - **The Prize:** The Top 3 Teams receive a **+10% Bonus** on their final course grade.
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2.0 THE 5 PROJECT OPTIONS

Students must select ONE of the following.

PROJECT A: The "Predator" Tracking Turret

Description: A 2-Axis (Pan/Tilt) Servo mechanism that autonomously tracks a specific target (Face or Colored Object) in real-time. The system must center the target in the frame and "fire" (laser or simulated sound) when locked on.

- **Computer Engineering Focus:** PID Control Loops (tuning proportional speed), Latency minimization (FPS vs. Servo Speed), Serial buffer management.
- **Required Hardware:**
 1. Raspberry Pi 4/5 + USB Webcam.
 2. Arduino Uno R4 (or Pico W).
 3. **Pan/Tilt Bracket Kit** (or 3D printed).

4. **2x Servos:** MG996R (High Torque) recommended, or SG90 (Micro).
5. **Laser Diode Module** (5V) or Piezo Buzzer.
6. External 5V/2A Power Supply for Servos (Critical).

PROJECT B: The "Quality Control" Conveyor Bot

Description: A system that visually inspects objects moving on a simulated production line. It must classify objects (e.g., "Red vs Green" or "Cube vs Sphere") and physically sort them into different bins using a servo arm or kicker.

- **Computer Engineering Focus:** State Machine Design (Idle -> Detect -> Classify -> Actuate -> Reset), Timing physics (accounting for conveyor speed), Multi-threading (Vision thread vs. Sorting thread).
- **Required Hardware:**
 1. Raspberry Pi 4/5 + USB Webcam.
 2. Arduino Uno R4 (or Pico W).
 3. **Conveyor Belt Kit** (or DIY using DC Motor + Cardboard/Fabric).
 4. **L298N Motor Driver** (for conveyor DC motor).
 5. **1x Servo (SG90)** for the "Kicker" arm.
 6. Objects to sort (e.g., colored candies, LEGO bricks).

PROJECT C: The "Tele-Surgeon" Robotic Mimic

Description: A robotic gripper/arm that mimics the user's hand movements in real-time. Using MediaPipe, the user's index/thumb pinch controls the robot's gripper, and hand elevation controls the arm height.

- **Computer Engineering Focus:** Data Normalization (Mapping 640px screen space to 180-degree servo space), Signal Filtering (Implementing Moving Average Filters to remove jitter), Inverse Kinematics basics.
- **Required Hardware:**
 1. Raspberry Pi 4/5 + USB Webcam.
 2. Arduino Uno R4 (or Pico W).
 3. **Robotic Gripper Kit** (Acrylic or Metal).

4. **2x to 3x Servos** (MG996R or SG90 depending on arm weight).
5. External Power Supply (Crucial for multi-servo stability).

PROJECT D: The Autonomous "Follow-Me" Rover

Description: A wheeled robot chassis that identifies a specific target (e.g., a yellow ball or a specific QR code) and drives toward it, maintaining a fixed distance (e.g., 20cm). It must turn left/right to keep the target centered.

- **Computer Engineering Focus:** Closed-loop control systems, PWM Motor duty cycle calculation, wireless architecture (Pi on the robot, powered by battery bank).
- **Required Hardware:**
 1. Raspberry Pi 4/5 + USB Webcam.
 2. Arduino Uno R4 (or Pico W).
 3. **2WD or 4WD Robot Chassis Kit.**
 4. **L298N Motor Driver.**
 5. **USB Power Bank** (for Pi) + **AA Battery Pack** (for Motors).

PROJECT E: The Biometric "Air-Pass" Security System

Description: A 2-Factor Authentication door lock.

- **Step 1:** Face Detection (User must be present).
- **Step 2:** Gesture Sequence (User must perform a specific "Passcode" in the air, e.g., "Fist -> Peace -> Open").
- **Action:** If successful, a servo unlocks a physical latch and an OLED displays "ACCESS GRANTED".
- **Computer Engineering Focus:** Sequence Logic/Stack implementation, Security timeouts, False-positive rejection logic, I2C Display interfacing.
- **Required Hardware:**
 1. Raspberry Pi 4/5 + USB Webcam.
 2. Arduino Uno R4 (or Pico W).
 3. **1x Servo (SG90)** (The Lock).
 4. **SSD1306 OLED Display** (I2C).

5. Red/Green LEDs (Status).

3.0 MANDATORY LEARNING PATH (Paul McWhorter)

To succeed, students must watch these specific lessons alongside the lectures.

Phase 1: The Muscle (Arduino/Hardware)

- *For Arduino Users:* New Arduino R4 WiFi Lessons **1, 4, 11 (PWM), 42 (Transistors)**.
- *For Pico Users:* Raspberry Pi Pico W Lessons **1, 6, 9 (PWM), 29 (Threads)**.

Phase 2: The Brain (Python/OpenCV on Pi)

- *Prerequisites:* Linux Lessons **1-5** (Navigating the Pi).
- *AI Core:* AI for Everyone (MediaPipe) Lessons:
 - **Lesson 1:** Setting up OpenCV.
 - **Lesson 12 & 13:** Color Detection & Tracking (For Projects A, B, D).
 - **Lesson 18:** Hand Tracking (For Projects C, E).
 - **Lesson 21:** Gesture Detection Logic.
 - **Lesson 26:** Face Detection (For Projects A, E).

Phase 3: The Bridge (Integration)

- *Crucial for ALL Projects:* **AI Lesson 29 (Serial Communication between Python and Arduino)**. This is the most critical video for the final project integration.
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4.0 FINAL REPORT: JOURNAL PAPER FORMAT

Students must submit a professional technical report following the structure of a top-tier IEEE/Elsevier journal.

Length: 10 to 20 Pages (Maximum).

Format: Double Column, Single Spaced.

Recommended Templates (Choose One):

1. **IEEE Transactions Template:** The gold standard for Computer Engineering.

- *Why:* Forces strict adherence to "Abstract, Methodology, Results, Conclusion."
- *Link:* [IEEE Template Selector](#) (Select "Transactions").

2. MDPI "Sensors" Journal Template:

- *Why:* Slightly more flexible, excellent for applied sensor/robotics projects.
- *Link:* [MDPI Latex/Word Templates](#).

Report Sections Required:

1. **Abstract:** 150-word summary of the system and results.
2. **Introduction:** Problem statement and why Embedded systems are relevant.
3. **System Architecture:** Block diagrams showing the Pi-to-Arduino data flow.
4. **Methodology:**
 - **Vision Algorithm:** How did you process the image? (HSV, Thresholds, MediaPipe).
 - **Control Logic:** Flowcharts of the State Machine.
 - **Communication:** How the Serial Packet is structured.
5. **Experimental Results:**
 - **Latency Graph:** Time (ms) from "Motion" to "Servo Response".
 - **Accuracy:** Success rate (e.g., "Sorted 45/50 balls correctly").
6. **Discussion:** Challenges faced (Lighting, Power issues, Serial lag) and how they were solved.
7. **Conclusion & Future Work.**
8. **References:** Datasheets, Paul McWhorter videos, and Library documentation.

5.0 EVALUATION CRITERIA (The Grading Rubric)

Total: 40% of Course Grade

1. **Live Demonstration (20%):**
 - Does it work? (Pass/Fail).

- Is it robust? (Does it crash if I wave my hand too fast?).
- **The "Unplug Test":** If the camera is unplugged and plugged back in, does the system recover or crash?

2. **Complexity & Engineering (10%):**

- Did they implement threading?
- Is the movement smooth (Filtered) or jittery (Raw)?
- Is the wiring neat or a mess?

3. **The Journal Paper (10%):**

- Adherence to IEEE/MDPI format.
- Quality of English and technical writing.
- Inclusion of data/graphs (Engineering analysis, not just a diary).

6.0 VIDEO DELIVERABLE

Deadline: End of Week 14.

Spec: 30 to 60 Seconds. Landscape Mode.

Content:

1. Title Card (Project Name + Student Names).
2. The Problem (1 sentence).
3. The Solution (Demo of the robot working).
4. Under-the-hood (Quick shot of the Pi/Arduino wiring and Python code).
5. *This video will be played during the voting session.*